ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 4 · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics practice exam 4.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, equivalent designs and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 C ✓ 1.1.3 B ✓ 1.1.4 B ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 C ✓ 1.2.2 J ✓ 1.2.3 I ✓ 1.2.4 F ✓ 1.2.5 E ✓ 1.2.6 G ✓ 1.2.7 H ✓ 1.2.8 D ✓ 1.2.9 B ✓ 1.2.10 A ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1.1 $R = (V_S - V_{LED}) \div I$ ✓ $= (6 - 3{,}2) \div 0{,}02 = 140\ \Omega$ ✓; choose $150\ \Omega$ ✓ (3)

2.1.2 $I_T = 20 + 20 = 40\ mA$ ✓ (1)

2.1.3 $P = I^2R$ ✓ $= 0{,}02^2 \times 140 = 0{,}06\ W$ ✓ (2)

2.2 Battery pack ✓, fuse and switch in series with the supply ✓, two parallel branches ✓ each with a resistor in series with an LED ✓, LEDs the right way round, neat with labels ✓ (5)

2.3 $V = 4 \times 1{,}5 = 6\ V$ ✓; capacity stays $2\ 000\ mAh$ ✓; $t = 2\ 000 \div 40$ ✓ $= 50\ h$ ✓ (4)

2.4 LEDs are never exactly the same; with one shared resistor the LED with the lowest voltage would take most of the current ✓, so it could burn out and the other one would be dim ✓. (2)

2.5 About 100 mA (a little above the normal 40 mA) ✓ so that it does not blow in normal use but blows on a short circuit. (Accept 63–250 mA with a reason.) (1)

2.6 Any TWO ✓✓: the longer leg is the anode; the flat side of the rim is the cathode; the larger plate inside is usually the cathode; test with the multimeter diode setting (it lights / conducts when red is on the anode). (2)

[20]

QUESTION 3

3.1 Connect the battery + and − to the long power rails ✓; place each resistor and LED so that they share a short row (the five holes in a row are connected) ✓; link each branch to the + and − rails with jumper wires ✓; keep the LED legs in different rows so they are not shorted, and check polarity ✓. (4)

3.2 In order ✓✓✓✓✓: clean the board and component leads; insert the component and bend the leads slightly; heat the pad and the lead together with a clean, tinned iron; feed solder onto the joint (not the iron) until it flows; remove the solder, then the iron, and let it cool without moving; trim the leads. (5)

3.3 Good: shiny, smooth, cone / volcano-shaped and covering the pad ✓. Dry: dull, grainy or a ball that does not wet the pad, may have cracks ✓. (2)

3.4 Setting ✓ and connection ✓ for each: (a) DC volts on a range above 6 V, probes across (in parallel with) the battery; (b) DC mA range, break the circuit and connect the meter in series with the LED; (c) ohms range, remove the resistor (or switch off the circuit) and put the probes across it. (6)

3.5 Any THREE ✓✓✓: LED in the wrong way round; dry joint or broken track; wrong resistor value (too high); faulty / blown LED; loose wire in the breadboard; flat battery or blown fuse (if both LEDs are off). (3)

[20]

QUESTION 4

4.1.1 $Y = T + A \cdot B$ ✓✓ (2)

4.1.2 TAB → Y: 000→0, 001→0, 010→0, 011→1, 100→1, 101→1, 110→1, 111→1 (any consistent column order). 8 rows ✓; T = 1 rows all 1 ✓; 011 = 1 ✓; other T = 0 rows 0 ✓ (4)

4.1.3 AND gate with inputs A and B ✓; OR gate with inputs T and the AND output ✓; output labelled Y ✓; correct symbols, neat ✓ (4)

4.2 (a) 0111 ✓✓ (b) 3 → 0011, 9 → 1001, so 0011 1001 ✓✓ (4)

4.3 Any TWO ✓✓: electronic switches have only two states (ON / OFF); two levels are easy to tell apart, so there are fewer errors from noise; simple, reliable circuits; easy to store and process. (2)

4.4 Triangle with a small circle at the output ✓; truth table 0→1, 1→0 ✓ (2)

4.5 Without it the input "floats" and can read random HIGH or LOW values from noise ✓; the pull-down resistor holds the input at 0 V (LOW) until the button connects it to the supply ✓. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF MARKING GUIDELINE